Worksheet 1: Section 1.1

Linear Equations and Rational Equations

Name: ______________________________ **Date:** ______________

Linear Equations

Linear equations are also known as equations of first degree. That is, variables have *no written exponents*. This generally excludes equations with variables in the denominators of fractions, since they would have an exponent of -1. However, we will see some equations with variables in their denominators that are *equivalent to* linear equations.

Solving Equations

To solve an equation means to find all possible solutions.

Good News! When solving an equation you can do anything you want—as long as you do it twice. Once to the whole left side and once to the whole right side.

Problem-Solving Strategy

Strategy for Solving Linear Equations:

1. **Distribute** to eliminate parentheses.
2. **Combine** "Like Terms."
3. Use **Addition/Subtraction** to "collect" variables on one side, constants on the other.
4. Use **Multiplication/Division** to isolate the variable.

Clearing Fractions

To clear fractions from an equation:

1. **Find the Lowest Common Denominator** (LCD) of ALL the fractions.
2. **Multiply** every term (on BOTH sides) by the LCD.
3. **Simplify** each term. There should be no fractions.
4. **Solve** the new equation using your algebra skills.

Variables in the Denominator

When variables appear in the denominator, we must identify *restricted values*. A restricted value makes any denominator equal to zero.

Steps:

1. Find the LCD of all fractions.
2. Identify any values that make the denominators zero (restricted values).
3. Multiply by the LCD.
4. Solve the resulting equation.
5. Check that the solution is not a restricted value.

Practice Problems

Show all work. Box your final answers.

Problem 1. Solve:

$$4x - 5 = 2(2x - x)$$

Answer: ______________________

Problem 2. Solve:

$$x - 4(x - 5) = 2 - (2 - x)$$

Answer: ____________________

Problem 3. Solve:

$$\frac{2x}{3} - \frac{5x}{6} - 3 = \frac{x}{2} - 5$$

Answer: ____________________

Problem 4. Solve:

$$\frac{9}{4} - \frac{1}{2x} = \frac{4}{x}$$

LCD: ____________________ Restricted Values: ____________________

Answer: ____________________

Problem 5. Solve:

$$\frac{1}{x-1} + \frac{x+1}{x^2+2x-3} = \frac{1}{x+3}$$

LCD: ______________________________ Restricted Values: ________________________

Answer: ________________________

Problem 6. (Challenge) Solve:

$$\frac{3}{x+2} - \frac{1}{x-1} = \frac{5}{x^2+x-2}$$

LCD: ______________________________ Restricted Values: ________________________

Answer: ________________________

Part 3: Answer Key

Answer Key

Problem 1: $x = 5$

Problem 2: $x = 5$

Problem 3: $x = 3$

Problem 4: LCD $= 4x$, Restricted Values: $x \neq 0$, $x = 2$

Problem 5: LCD $= (x-1)(x+3)$, Restricted Values: $x \neq 1, -3$, $x = -5$

Problem 6: LCD $= (x+2)(x-1)$, Restricted Values: $x \neq -2, 1$, $x = 3$